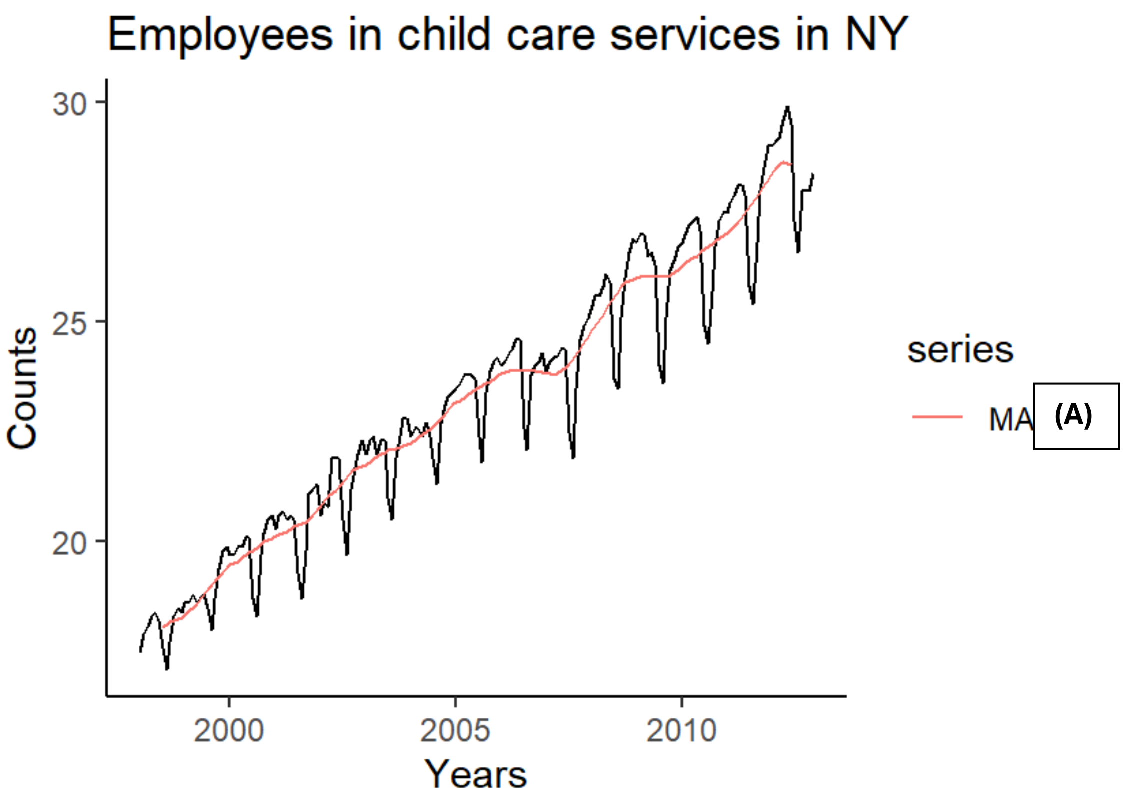
STA202F 2026: Timer Series Workshop

Question 1:

The time series under investigation contains information on the number of employees (measured every month) in child day care services in New York City, between January 1998 and December 2012. Assess the information and answer the questions that follow.

1. Discuss the (observed and unobserved) time series components in the time plot below. (3)



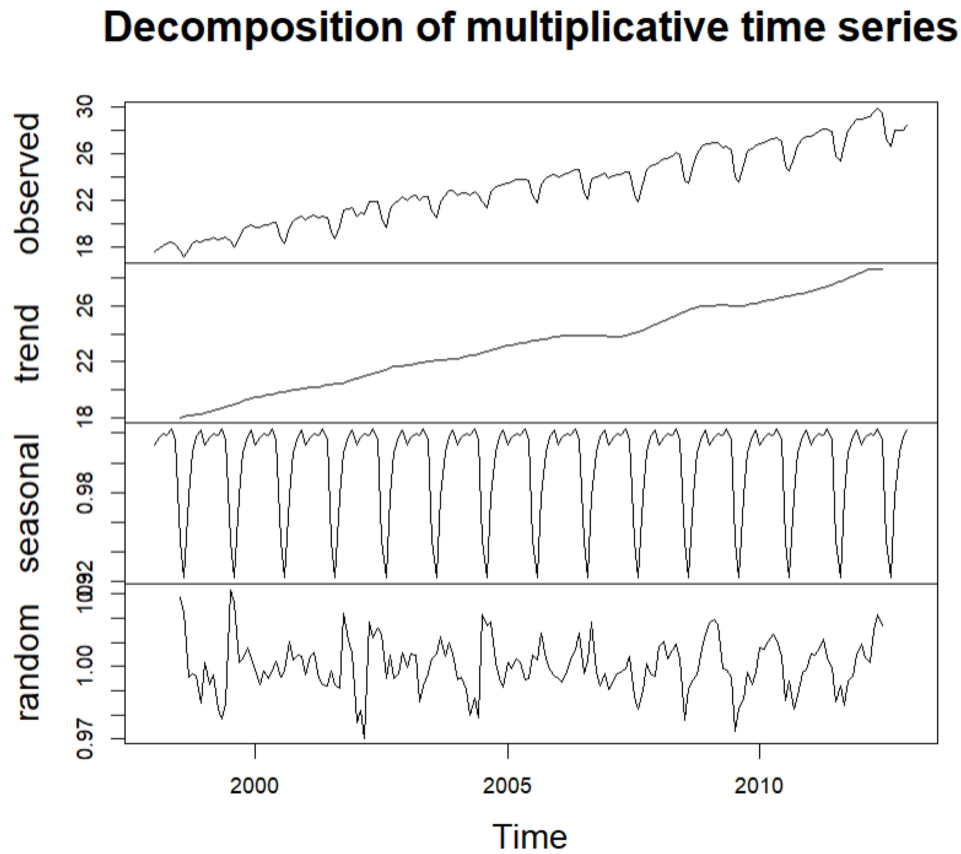
2. Explain the concept of moving average smoothing and centred moving average smoothing. (2) Why would we consider it for this time series? (1) What should the value of **A**, in the time plot, be? (0.5) Discuss one drawback of using the moving average smoothing technique on a time series, and point out whether you think this drawback is critical for the specific time series under consideration. (2)

Classical decomposition sheds some light on detrended and deseasonalised data.

3. Investigate the output on the classical decomposition of the time series under consideration. You may assume that the first adjusted seasonal index is associated with January. Calculate the value of **B** and interpret this value. (3)

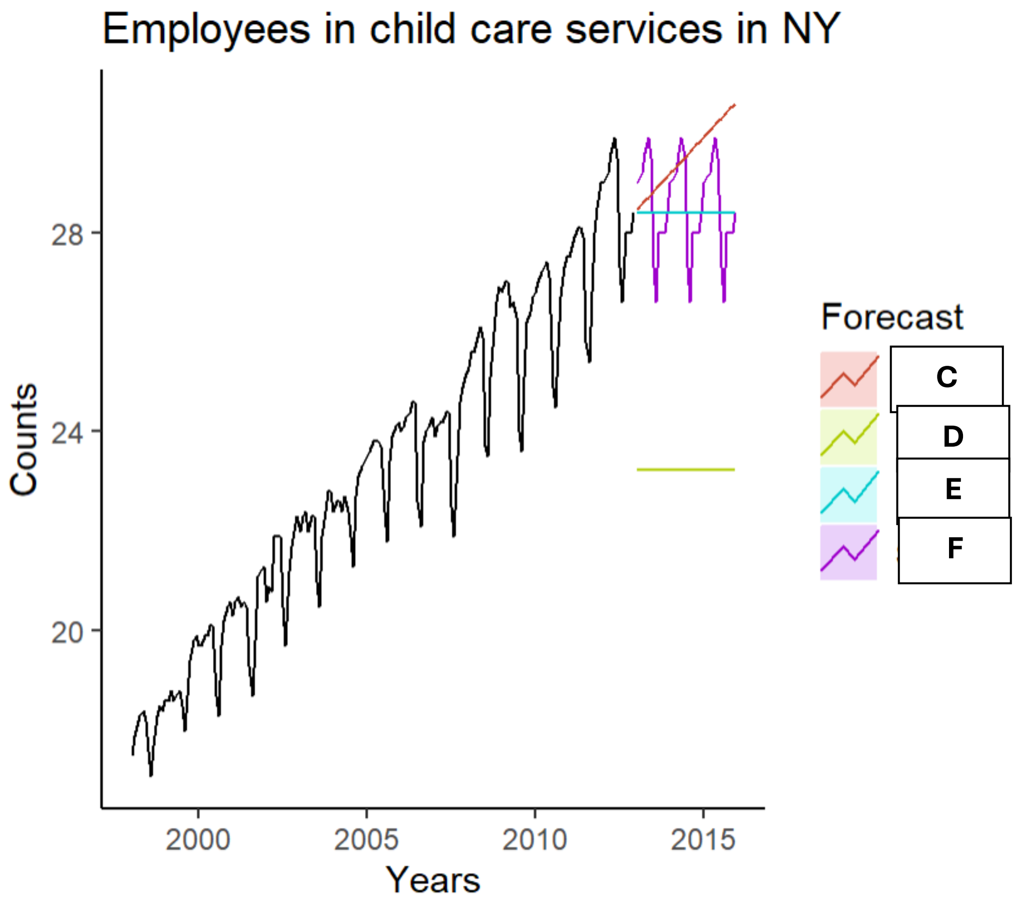
```
$figure
 [1] 1.0122264 1.0173918 1.0194796 1.0184221 1.0225669 1.0150052
 [7] 0.9479798     B     0.9777202 1.0072128 1.0176673 1.0215753
```

4. Investigate the output on the classical decomposition of the time series under consideration. Do you think that a linear regression model would fit the deseasonalised observations well? Explain your reasoning from the available output. (2) Write down the classical decomposition forecasting model and calculate the forecast for t = 192. You may assume that the 192nd month was December. (2)




```
Call:
lm(formula = trend ~ time)

Residuals:
     Min       1Q   Median       3Q      Max 
-0.71027 -0.19520  0.05069  0.19222  0.56221 

Coefficients:
             Estimate Std. Error t value Pr(>|t|)    
(Intercept) 1.788e+01  4.176e-02     428   <2e-16 ***
time        5.938e-02  4.068e-04     146   <2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 0.2557 on 166 degrees of freedom
Multiple R-squared:  0.9923,	Adjusted R-squared:  0.9922 
F-statistic: 2.131e+04 on 1 and 166 DF,  p-value: < 2.2e-16
```

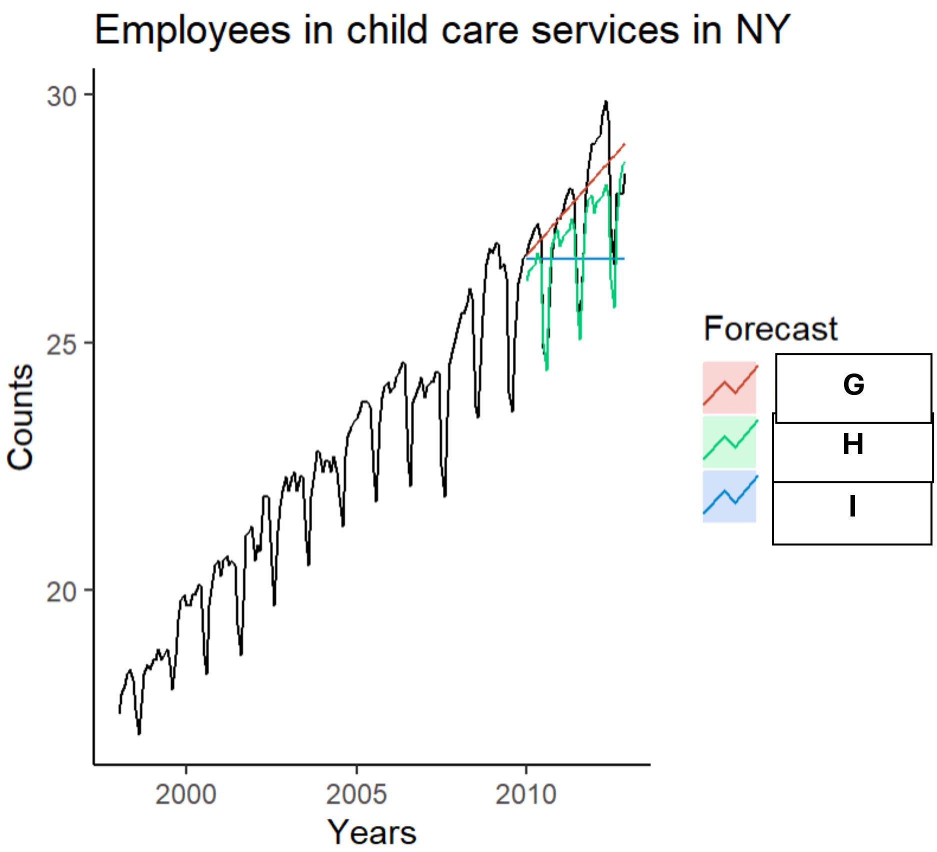
Forecasting is an important aspect of time series. Investigate the simple forecasting methods fit to the data below.



5. Explain which simple forecasting method you would expect to forecast short-term observations the best. Why is this the best approach? Which letter in the forecasting plot best represents your choice? How would you code this forecasting technique? (3)
6. Explain which simple forecasting method you would expect to forecast long-term observations the best. Why is this the best approach? Which letter in the forecasting plot best represents your choice? What code would you use to obtain this forecasting technique? (3)

Exponential smoothing methods are more sophisticated than the simple forecasting methods described above. Investigate the output below and answer the questions that follow.

7. Identify the exponential smoothing models indicated as **G, H,** and **I** on the forecasting plot. (1.5) Discuss the suitability of each for this particular time series. (3)



8. Write down the expression of the trend equation of the model **H**, using the initial values, so that you might obtain b_1. You may assume l_1 is 20.38. (2)
9. Calculate the values of **J** and **K,** the upper bound of the 80% prediction interval and the lower bound of the 95% prediction interval, respectively. Use the appropriate standard error (sigma) from the respective models. (4)
10. Write down the smallest mean absolute error between the three fitted models. Why are we interested in the smallest possible MAE? (2)
11. Calculate the smallest mean squared error between the three fitted models.(1) What happens to this measure when we have outliers in our time series? (1) Explain one way in which outliers can be handled in a time series. (1)
12. What is the significance of the large alpha-value in Model I? (1)
13. Which other measure can we use to decide on which model fits the time series the best? Explain the function of this measure; how does it work, and what does it mean? (2)

OUTPUT for Exponential smoothing models

Model I:

```
Forecast method: 

Model Information:

Call:
ses(y = train, h = 36)

  Smoothing parameters:
    alpha = 0.9999

  Initial states:
    l = 17.4993

  sigma:  0.6857

     AIC     AICc      BIC
610.9650 611.1365 619.8745

 Forecasts:
          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
 Jan 2010       26.69997 25.82123          25.35606 28.04388
 Feb 2010       26.69997 25.45731 27.94263 24.79948 28.60046
 Mar 2010       26.69997 25.17805 28.22189 24.37240 29.02754
 Apr 2010       26.69997 24.94263 28.45731 24.01235 29.38759
 May 2010       26.69997 24.73521 28.66473 23.69513 29.70481
 Jun 2010       26.69997 24.54769 28.85225 23.40834 29.99160
 Jul 2010       26.69997 24.37525 29.02469 23.14461 30.25533
 Aug 2010       26.69997 24.21474 29.18520 22.89914 30.50080
 Sep 2010       26.69997 24.06399 29.33595 22.66859 30.73135


                     ME      RMSE      MAE      MPE
Training set 0.06389978 0.6809043 0.454198 0.246246
Test set     0.87503001 1.5341655 1.280576 2.965889
                 MAPE      MASE      ACF1 Theil's U
Training set 2.073693 0.6007428 0.2034256        NA
Test set     4.581516 1.6937472 0.7741615  1.792601
```

Model G:

```
Forecast method: G

Model Information:
G

Call:
holt(y = train, h = 36)

  Smoothing parameters:
    alpha = 0.9999
    beta  = 1e-04

  Initial states:
    l = 17.4487
    b = 0.0649

  sigma:  0.6875

     AIC     AICc      BIC
613.6974 614.1322 628.5465
```

Forecasts:

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
Jan 2010	26.76491	25.88380	27.64601	K	28.11244
Feb 2010	26.82984	25.58377	28.07591	24.92414	28.73554
Mar 2010	26.89478	25.36861	28.42095	24.56070	29.22885
Apr 2010	26.95971	25.19737	28.72205	24.26444	29.65498
May 2010	27.02464	25.05419	28.99509	24.01110	30.03818
Jun 2010	27.08958	24.93096	29.24819	23.78825	30.39090
Jul 2010	27.15451	24.82282	29.48619	23.58850	30.72051
Aug 2010	27.21944	24.72665	29.71224	23.40704	31.03184
Sep 2010	27.28437	24.64024	29.92851	23.24051	31.32824

	ME	RMSE	MAE	MPE
Training set	-0.0006979817	0.6779138	0.4448250	-0.04896107
Test set	-0.3262399606	1.1108790	0.8094174	-1.33934827

	MAPE	MASE	ACF1	Theil's U
Training set	2.035542	0.5883457	0.2036955	NA
Test set	3.012743	1.0705720	0.6956806	1.372577

Model H:

```
Forecast method: [ H ]

Model Information:
[ H ]
Call:
hw(y = train, h = 36, seasonal = "multiplicative")

  Smoothing parameters:
    alpha = 0.5314
    beta  = 0.0052
    gamma = 0.0343

  Initial states:
    l = 18.684
    b = 0.072
    s = 1.0219 1.0213 1.0151 0.987 0.9255 0.9469
           1.0101 1.021 1.0154 1.0138 1.0136 1.0084

  sigma:  0.0146

     AIC     AICc      BIC
406.1620 411.0192 456.6488

                      ME      RMSE       MAE        MPE
Training set -0.01998401 0.2912867 0.2016162 -0.1146766
Test set      0.54445126 0.7873714 0.6504335  1.9184605
                  MAPE      MASE      ACF1 Theil's U
Training set 0.9291897 0.2666667 0.2740832        NA
Test set     2.3080726 0.8602928 0.8335591 0.9012322
```

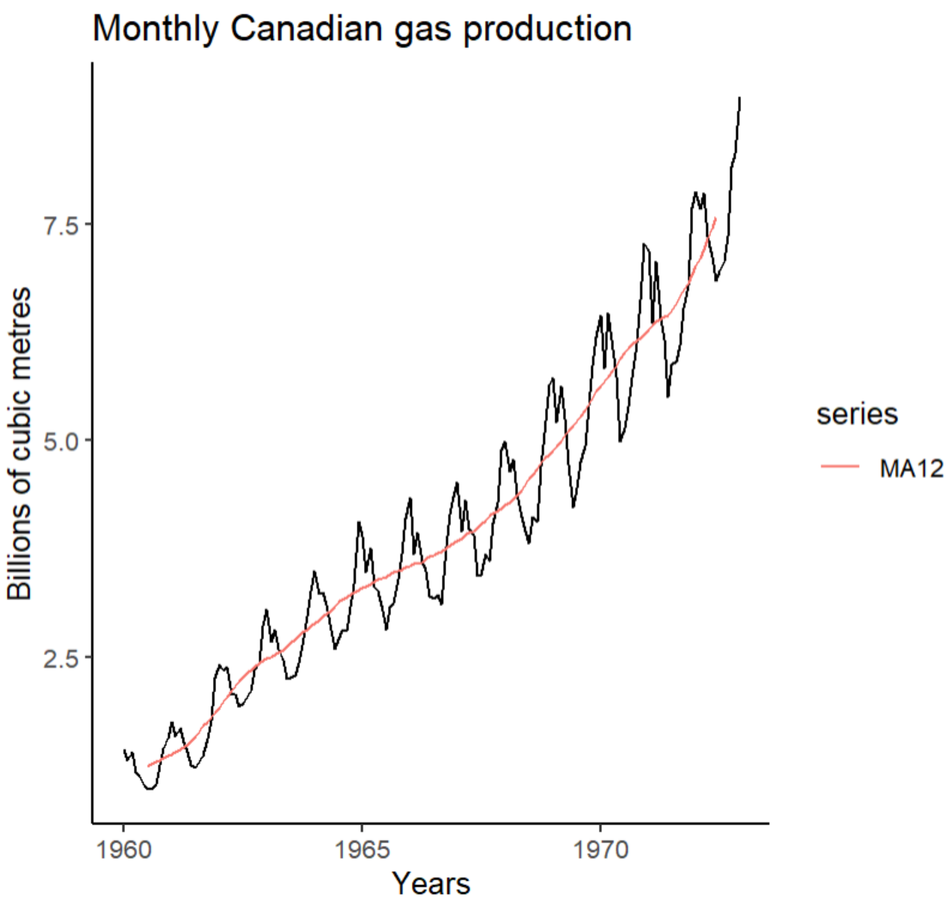
What did I not include in these questions? TABLES!! Practice ON YOUR OWN how to calculate:

- specific moving averages/centred moving averages in a table.
- detrended data,
- deseasonalized data,
- adjusted seasonal indices,
- using the forecast equations to obtain forecasted values, etc.
- KPSS test to assess stationarity of the mean – you need to know how to code and interpret this

<u>Question 2:</u> Technically, the following time series contains seasonality and our non-seasonal ARIMA models are not a good idea to use here.

The time series under investigation contains information on monthly gas production in Canada, measured between January 1960 and December 1972. Assess the information and answer the questions that follow.

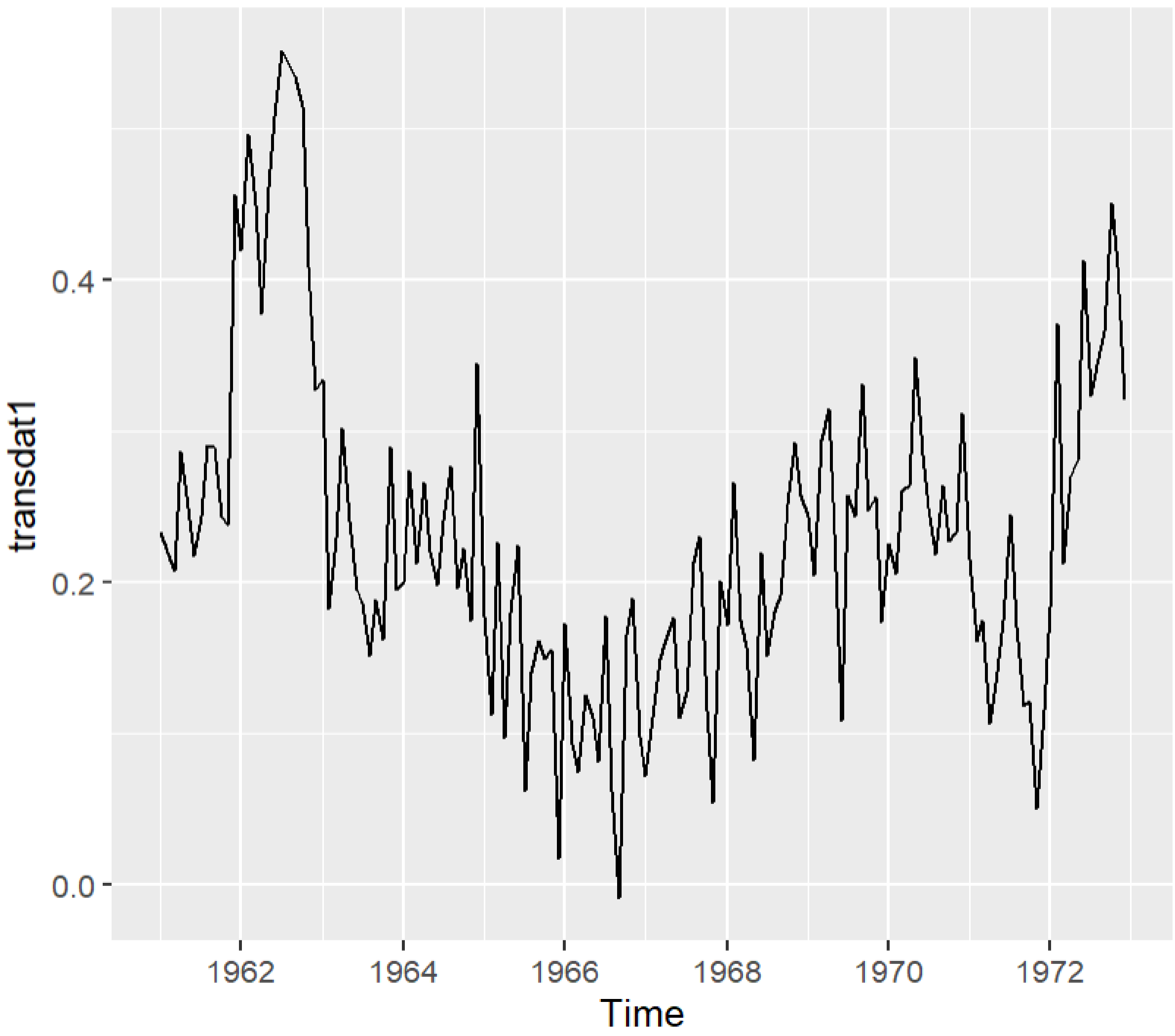
1. Discuss the (observed and unobserved) time series components in the time plot below. (3)

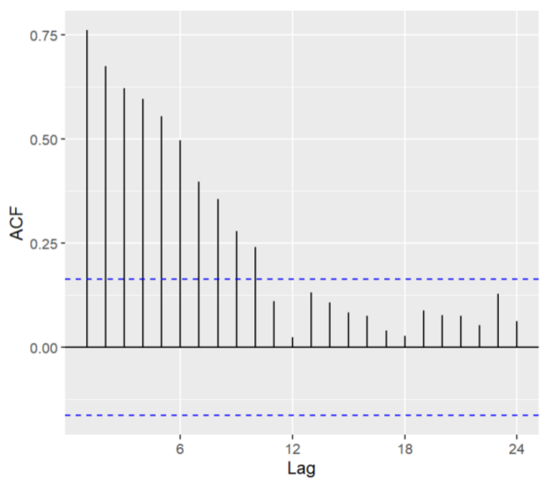


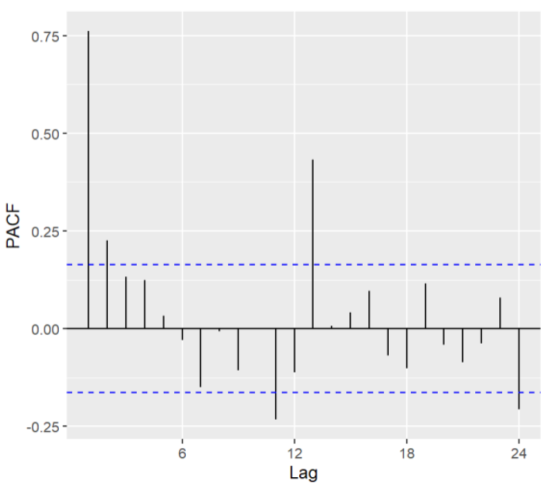
2. Is this time series stationary? Explain your reasoning in as much detail as possible and explain how you would go about making this time series stationary, if necessary. (3)
3. Investigate the lambda-value generated by the Box-Cox function in R Studio. What does this value tell us? (2) Would a square-root or cube-root transformation of the time series have a similar effect? (1)

```
> lambda <- BoxCox.lambda(data)
> lambda
[1] 0.3503923
```

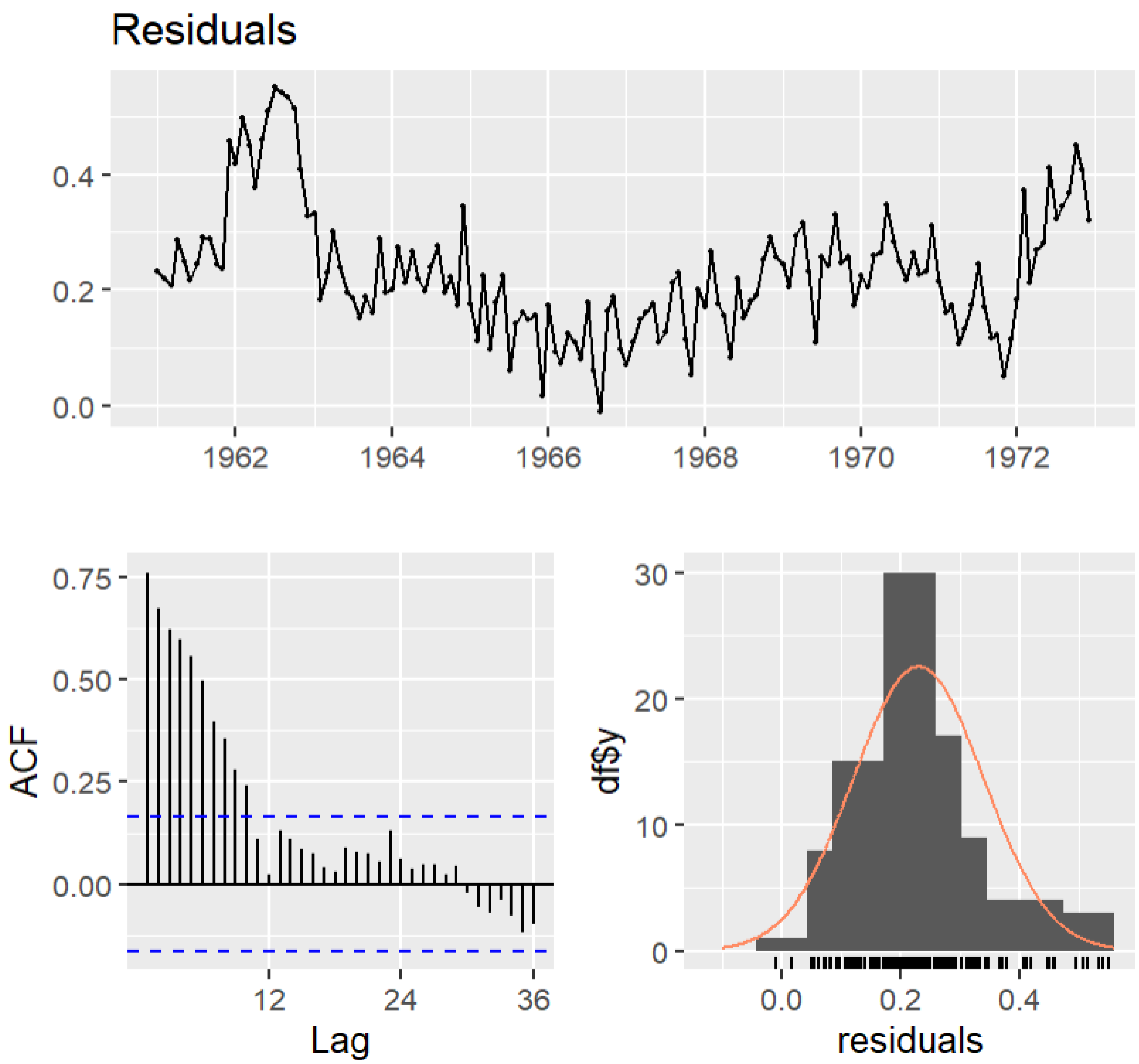
4. Applying first-order differencing at lag g=**L,** to address seasonality, together with stabilising the variation, leads to the plot of following transformed time series (called *transdat1*). Write down the value of **L** (0.5). Also consider the ACF plot of this transformed time series. Is this time series a white noise process? Why or why not? (1.5)





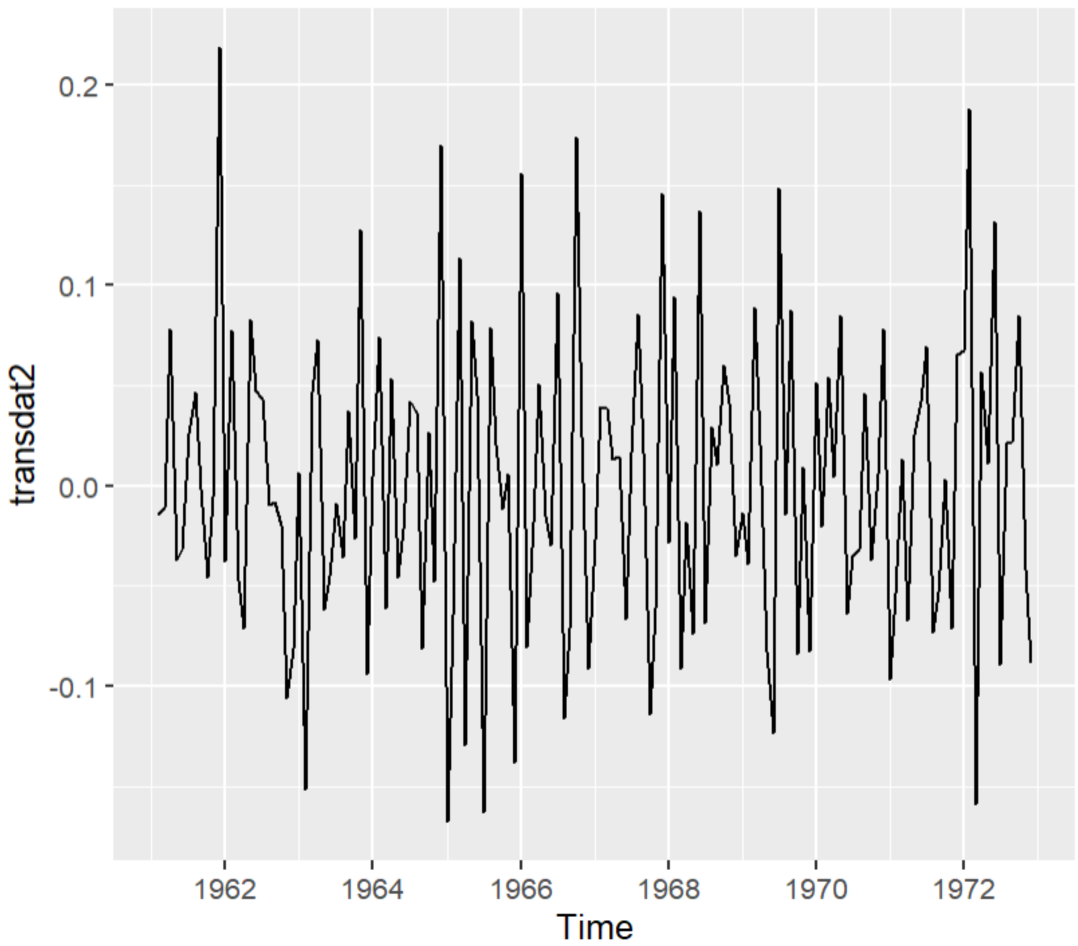


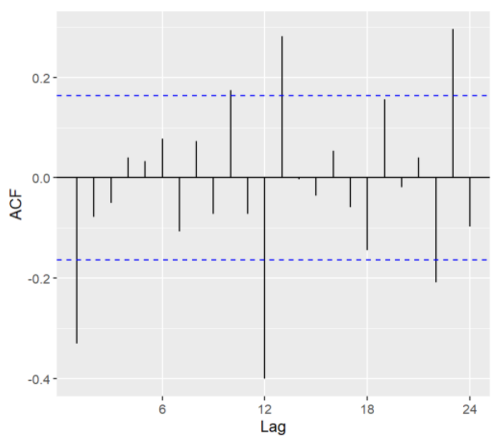
5. The ACF and PACF of *transdat1* were given. What ARIMA model would you intuitively think to fit to *transdat1*? (1) Why do you think this model could be appropriate? (1)
6. The following output was generated for the residuals of *transdat1*. Investigate the residuals and explain your findings in as much detail as possible. (4) Can we conclude that model *transdat1* has captured all the underlying patterns/information in the time series? (1)

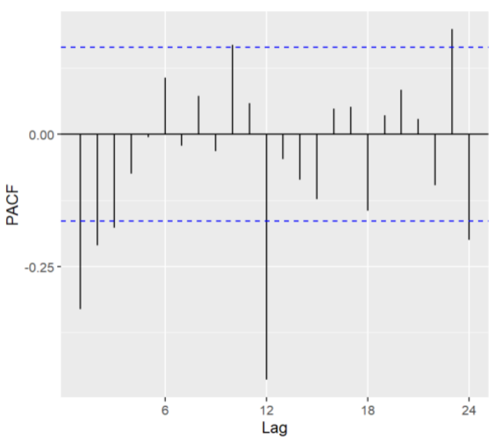



```
	Ljung-Box test

data:  Residuals
Q* = 429.43, df = 24, p-value < 2.2e-16

Model df: 0.   Total lags used: 24
```

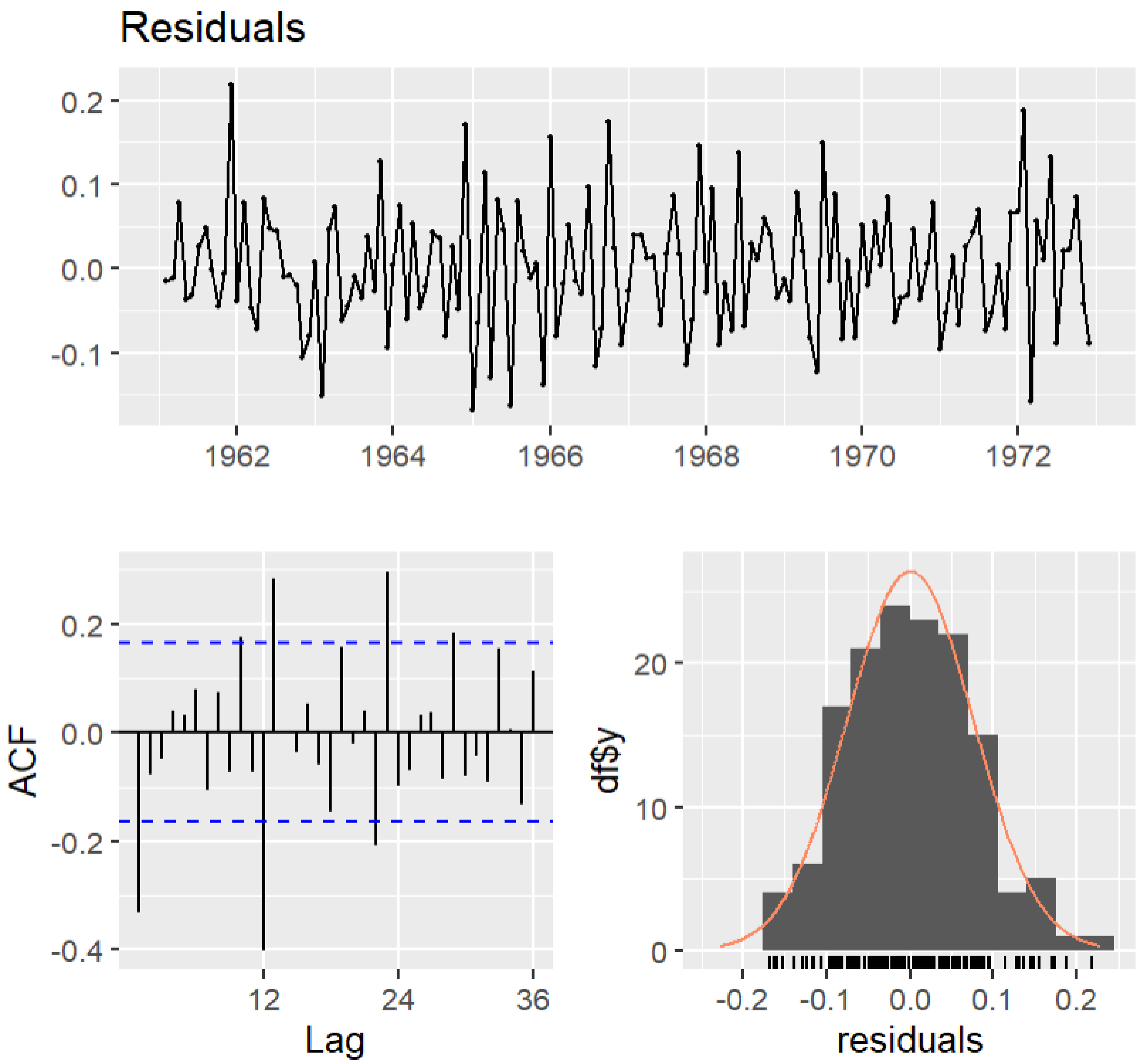
7. Applying second-order differencing, first at lag *g*=**L,** to address seasonality, and then at lag 1**,** together with stabilising the variation, leads to the following transformed time series (called *transdat2*). Assess the ACF of this transformed time series as well. Is this time series a white noise process? Why or why not? (1.5)







8. The ACF and PACF of *transdat2* were given. What ARIMA model would you intuitively think to fit to *transdat2*? (1) Why do you think this model could be appropriate? (1)
9. The following output was generated for the residuals of *transdat2*. Investigate the residuals and explain your findings in as much detail as possible. (4) Can we conclude that model *transdat2* has captured all the underlying patterns/information in the time series? (1)




```
        Ljung-Box test

data:  Residuals
Q* = 99.11, df = 24, p-value = 4.26e-11

Model df: 0.   Total lags used: 24
```

10. The following models were fit:

- ARIMA(2,1,0)on *transdat1* data
- ARIMA(3,2,1) on *transdat2* data
- ARIMA(2,2,2) on *transdat2* data
- ARIMA(2,2,1) on *transdat2* data

i) Write down the ARIMA(2,1,0) model with its fitted parameters and calculate the mean. (3)

ii) Calculate the AIC value, **M**, for the ARIMA(3,2,1) model.(2)

iii) Which ARIMA model proved to be the best out of the four fitted models? Explain your answer with an appropriate measure from the output.

```
Call:
arima(x = transdat1, order = (2,1,0)

Coefficients:
         ar1     ar2  intercept
      0.5776  0.2393     0.2342
s.e.  0.0805  0.0811     0.0302

sigma^2 estimated as 0.004739:  log likelihood = 180.53,  aic = -353.06
```

```
Call:
arima(x = transdat2, order = (3,2,1)

Coefficients:
          ar1      ar2      ar3      ma1  intercept
      -0.1997  -0.1897  -0.1291  -0.2520     0.0009
s.e.   0.3067   0.1495   0.1132   0.3026     0.0028

sigma^2 estimated as 0.004687:  log likelihood = 180.38,  aic = M
```

```
Call:
arima(x = transdat2, order = (2,2,2)

Coefficients:
         ar1      ar2      ma1     ma2  intercept
      1.1695  -0.5132  -1.6407  0.9032     0.0007
s.e.  0.1091   0.1072   0.0661  0.0743     0.0042

sigma^2 estimated as 0.004386:  log likelihood = 184.5,  aic = -357
```

```
Call:
arima(x = transdat2, order = (2,2,1)

Coefficients:
         ar1      ar2      ma1  intercept
      0.0361  -0.0884  -0.4833     0.0009
s.e.  0.1902   0.1140   0.1750     0.0028

sigma^2 estimated as 0.004724:  log likelihood = 179.84,  aic = -349.69
```

R code for generating the output in the above questions

```
library(ggplot2)
library(fpp2)
library(forecast)
library(urca)
library(fpp3)
library(tidyverse)

#Example 1
data <- fpp3::ny_childcare
print(data)
data <- as.ts(data)
data <- window(data, start = c(1998,1), end=c(2012,12))

autoplot(data, series = "Counts", col="black")+
  autolayer(ma(data,12), series = "MA12")+
  xlab("Years")+
  ylab("Counts")+
  ggtitle("Employees in child care services in NY")+
  theme_classic()

#Classic decompostion
decomp_model<-decompose(data, type = "multiplicative")
print(decomp_model)
plot(decomp_model)

#fit linear regression to trend:
trend <- na.omit(decomp_model$trend)
time <- 7:174 #first 6 and last 6 time points are missing
because of CMA(12)
model.lm <- lm(trend ~ time)
summary(model.lm)
model.lm$coefficients

#Simple forecasting methods:drift and seasonal naive
autoplot(data, series = "Counts", col="black")+
  autolayer(snaive(data, h=36), series = "Snaive", PI =
FALSE)+
  autolayer(rwf(data, drift=TRUE, h=36), series = "Drift", PI
= FALSE)+
  autolayer(rwf(data, h=36), series = "Naive", PI = FALSE)+
  autolayer(meanf(data, h=36), series = "Mean", PI = FALSE)+
  xlab("Years")+
```

```
  ylab("Counts")+
  ggtitle("Employees in child care services in NY")+
  theme_classic()+
  guides(colour=guide_legend(title="Forecast"))

#Forecasting accuracy: Exponential smoothing
train <- window(data, end=c(2009,12))
test <- window(data, start=c(2010,1))

fcses<-ses(train, h = 36)
summary(fcses)
accuracy(fcses, test)

fcholt<-holt(train, h=36)
summary(fcholt)
accuracy(fcholt, test)

fchw<-hw(train, seasonal = "multiplicative", h = 36)
summary(fchw)
accuracy(fchw, test)

autoplot(data)+
  autolayer(fcses, PI = FALSE, series = "SES")+
  autolayer(fcholt, PI = FALSE, series = "Holt")+
  autolayer(fchw, PI = FALSE, series = "Holt-Winters")+
  xlab("Years")+
  ylab("Counts")+
  ggtitle("Employees in child care services in NY")+
  theme_classic()+
  guides(colour=guide_legend(title="Forecast"))



#Example 2
data <- fpp3::canadian_gas
print(data)
data <- as.ts(data)
data <- window(data, start = c(1960,1), end=c(1972,12))

autoplot(data, series = "Counts", col = "black")+
  autolayer(ma(data,12), series = "MA12")+
  xlab("Years")+
  ylab("Billions of cubic metres")+
  ggtitle("Monthly Canadian gas production")+
```

```
  theme_classic()

#Transformations: stabilise variance and take second-order
differencing
lambda <- BoxCox.lambda(data)
lambda

transdat1 <- data %>% BoxCox(lambda) %>%diff(12)
ndiffs(transdat1)
autoplot(transdat1)
ggAcf(transdat1)
ggPacf(transdat1)
checkresiduals(transdat1)

transdat2 <-diff(transdat1)
ndiffs(transdat2)
autoplot(transdat2)
ggAcf(transdat2)
ggPacf(transdat2)
checkresiduals(transdat2)


#fitting different ARIMA models
trans1.ts<-arima(transdat1,order=c(2,0,0))
trans2.ts<-arima(transdat2, order = c(3,0,1))
trans3.ts <- arima(transdat2, order = c(2,0,2))
trans4.ts <- arima(transdat2, order = c(2,0,1))

summary(trans1.ts)
summary(trans2.ts)
summary(trans3.ts)
summary(trans4.ts)
```